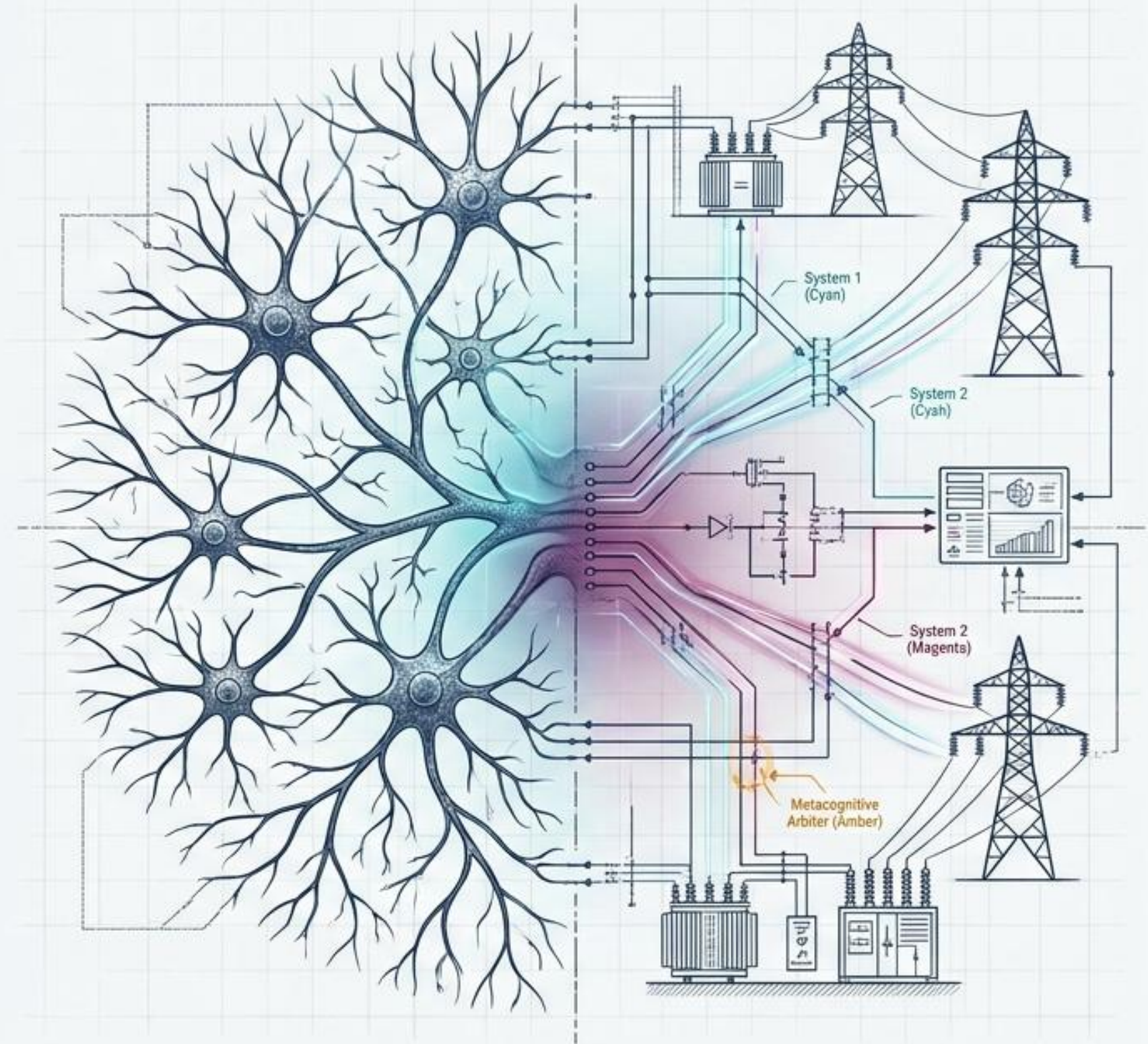


COGNITIVE ADAPTIVE POWER SYSTEM MANAGEMENT (CAPSM)

A Brain-Inspired Dual-Process AI
Framework for Modern Power Grids

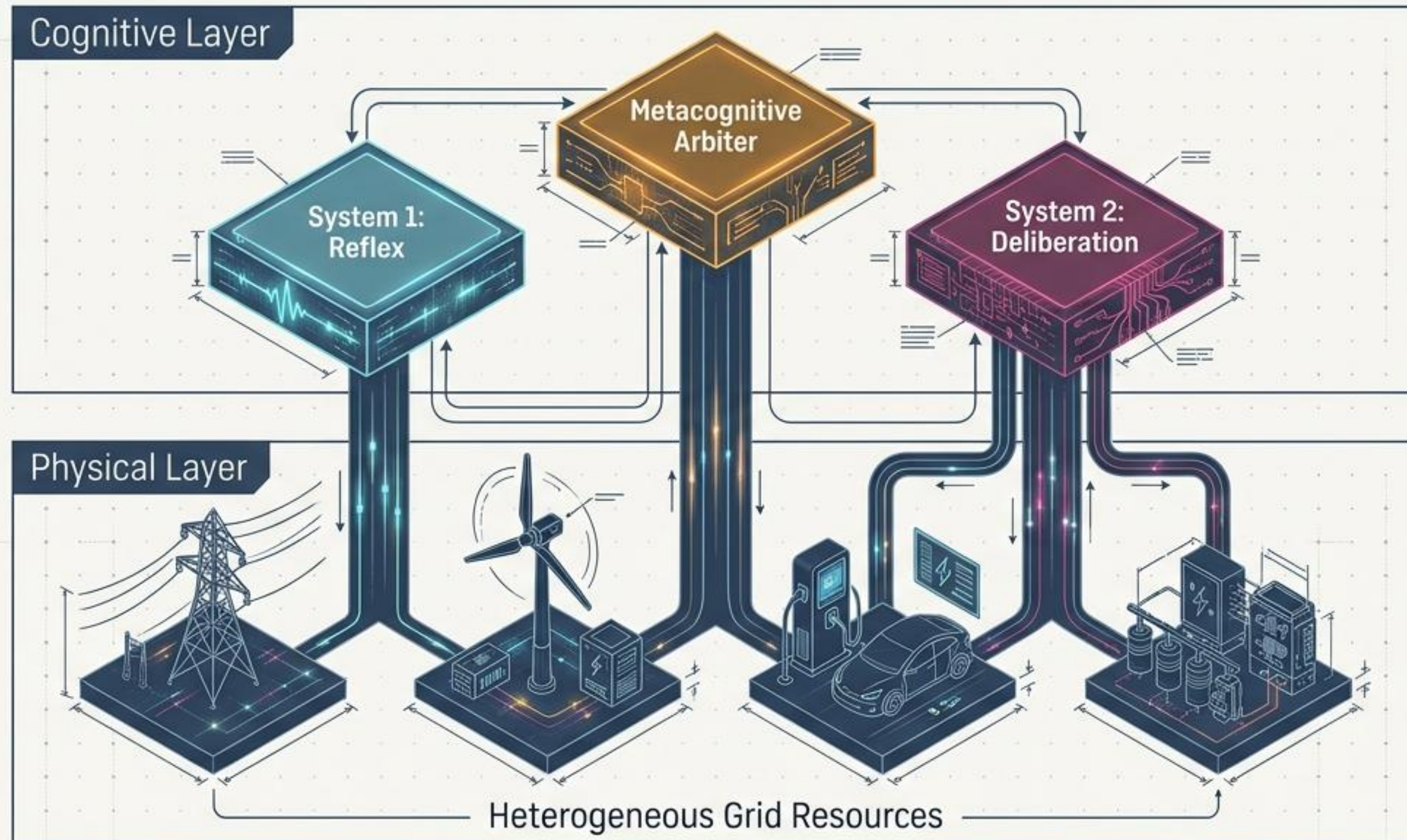
Mahmoud Kiasari | Dalhousie University

Validated via Hardware-in-the-Loop on
OPAL-RT Simulator



The Grid as a Biological Organism: CAPSM Architecture

Multi-timescale AI handles both split-second volatility and long-term planning.



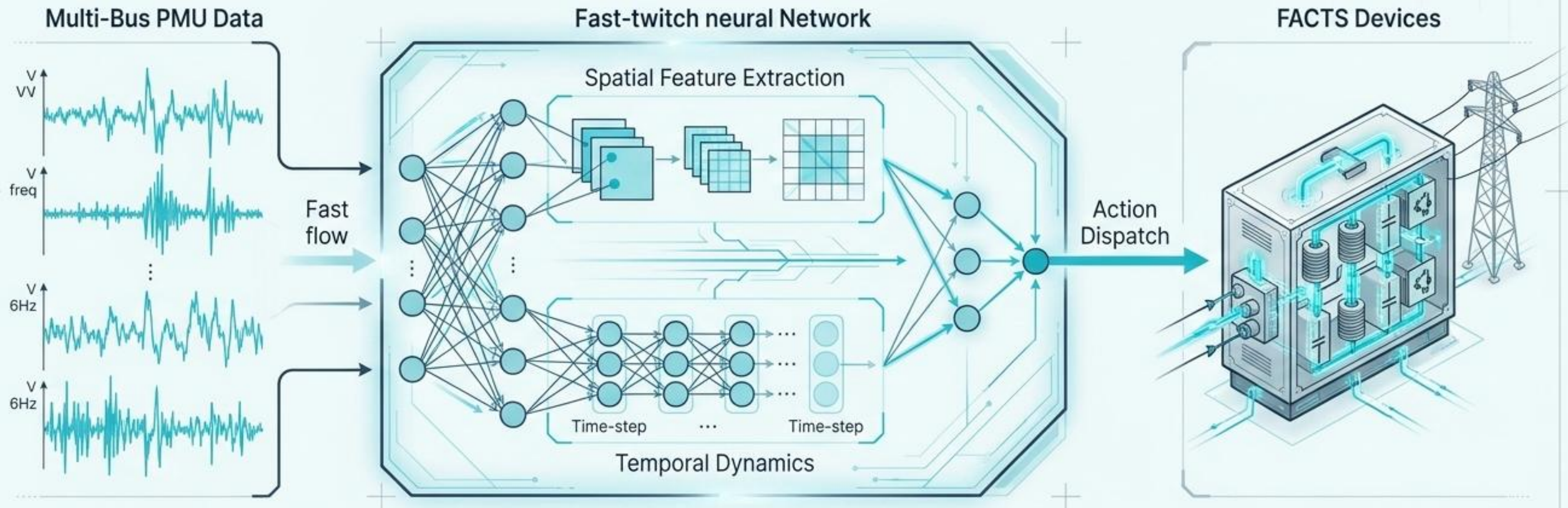
Translating Cognitive Dual-Process Theory to Grid Control

	System 1	System 2
Kahneman Model	 Fast, automatic, intuitive, and reflexive.	 Slow, analytical, logical, and deliberative.
Grid Application	 Reflexive Protection Controller Immediate stabilization and fault detection.	 Deliberative Optimizer Economic dispatch, V2G scheduling, and system efficiency.
Response Target	 Sub-5 milliseconds	 50 milliseconds to seconds

System 1 Reflexive Controller Overview

Goal: Immediate grid stabilization, fault detection, and oscillation damping.

Architecture: CNN-LSTM hybrid model processing time-series data.



System 1: CNN Architecture Details

Extracting spatial features from multi-bus PMU data.

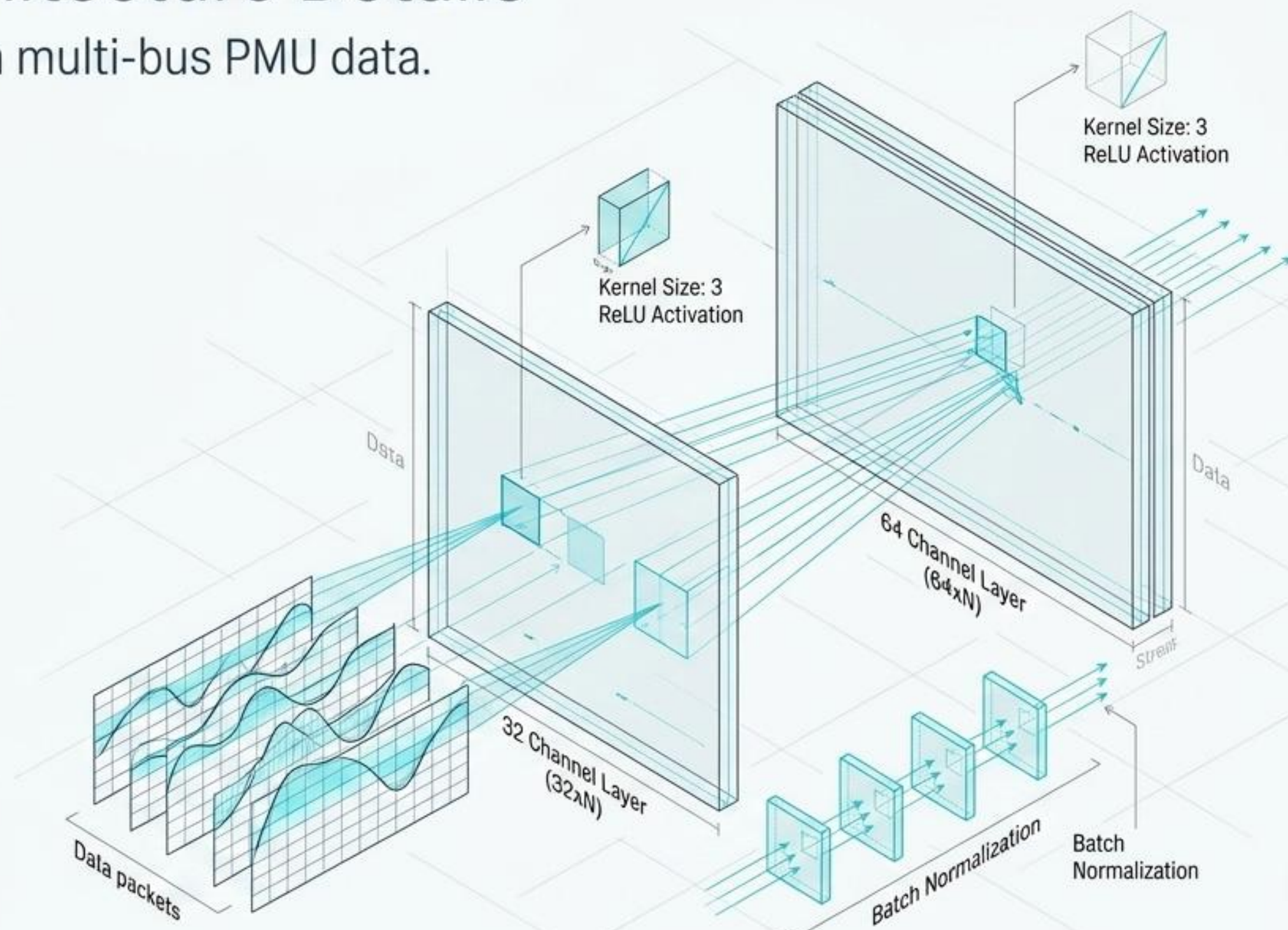
1D Convolutional Layers

32 and 64 Channels

Kernel Size: 3

ReLU Activation

Batch Normalization



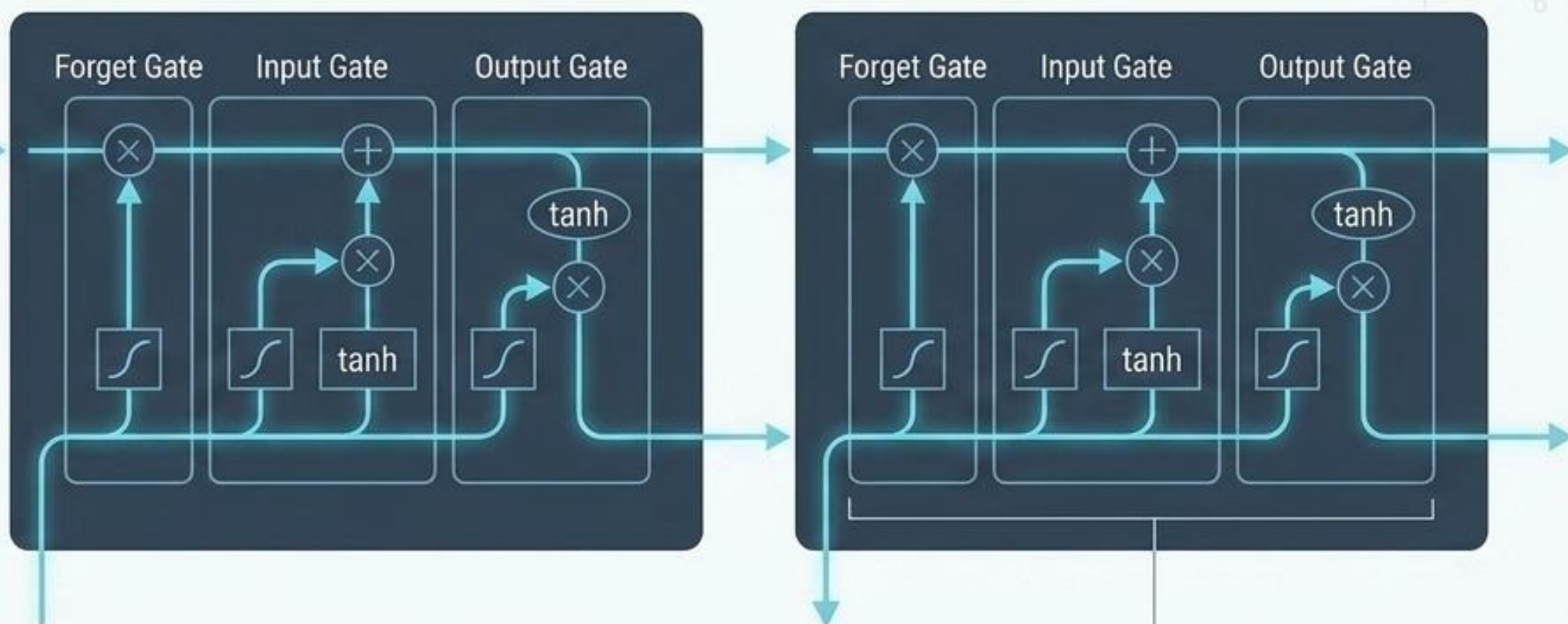
System 1: LSTM Architecture Details

Capturing temporal signatures like decaying oscillations and fault trajectories.

Flattened CNN Output

2 LSTM Layers

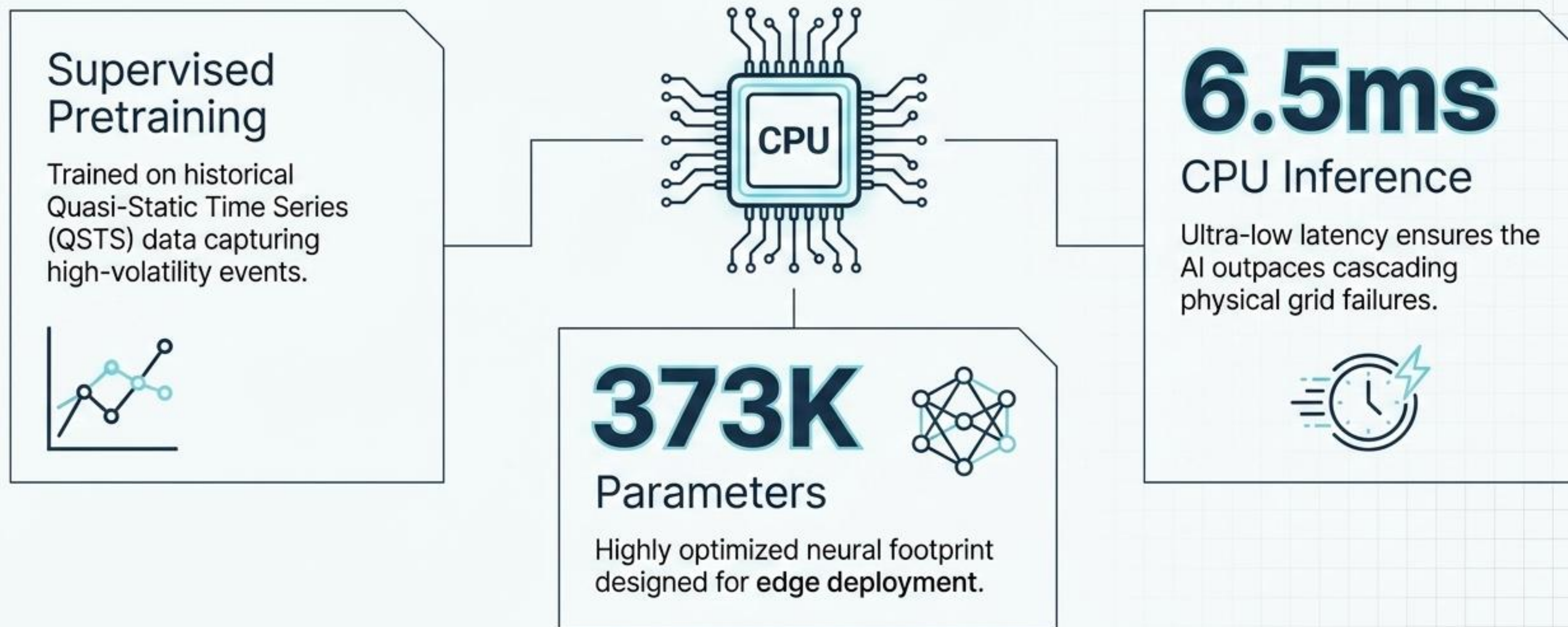
Temporal Memory Cells



Hidden Size: 128

System 1: Training Methodology & Sub-Millisecond Execution

Training Methodology & Sub-Millisecond Execution

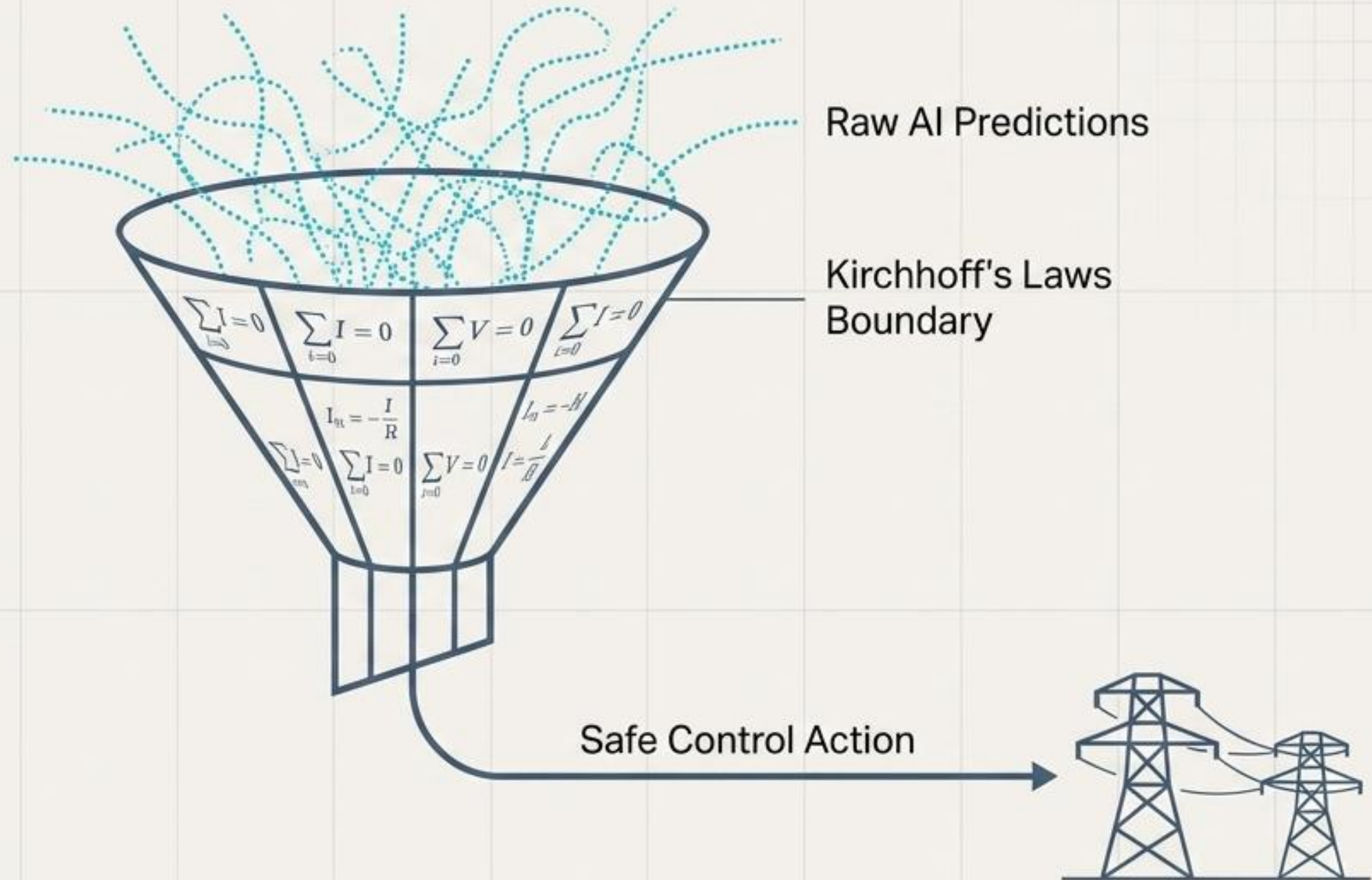


Physics-Informed Neural Network (PINN) Safety Filter

Problem: Raw neural networks can hallucinate physically impossible actions.

Filter: Control actions are strictly clipped to safe operating envelopes before execution.

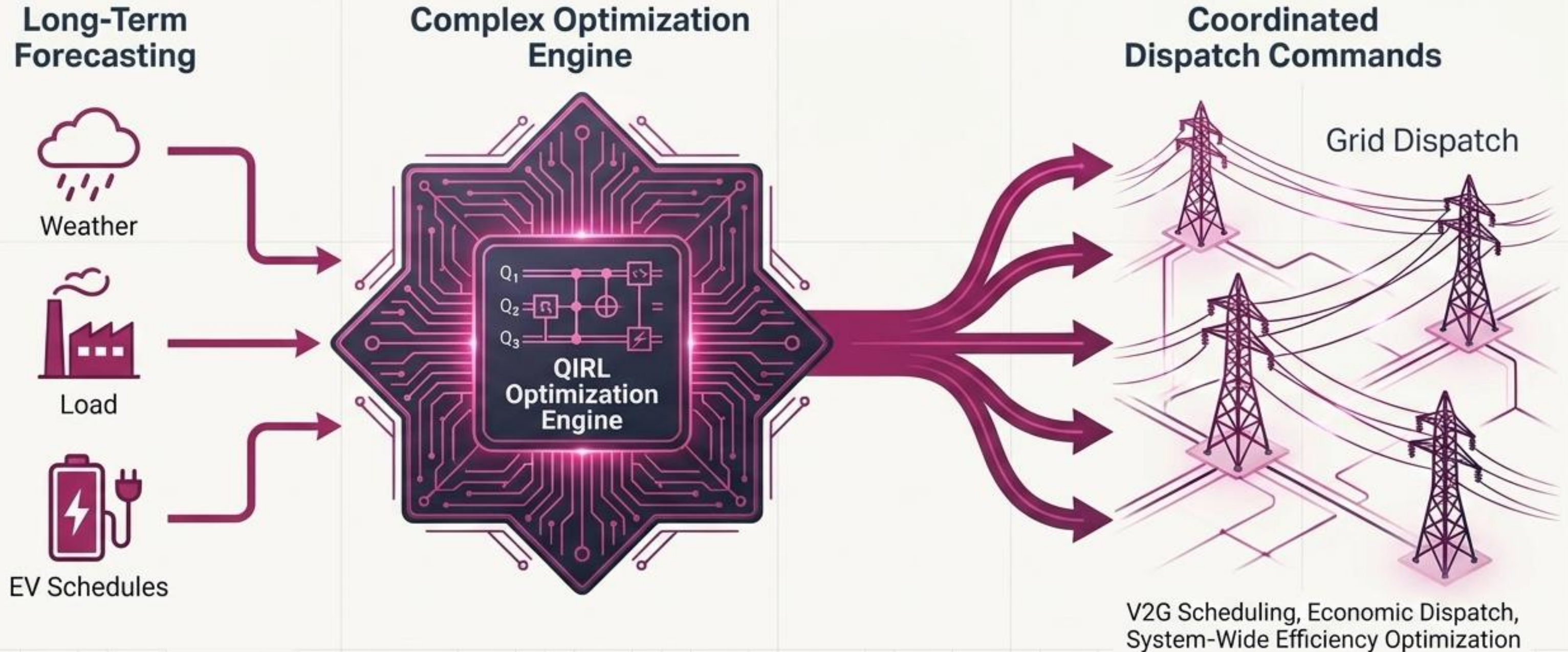
Target: 100% constraint compliance.



System 2: QIRL Deliberative Controller Overview

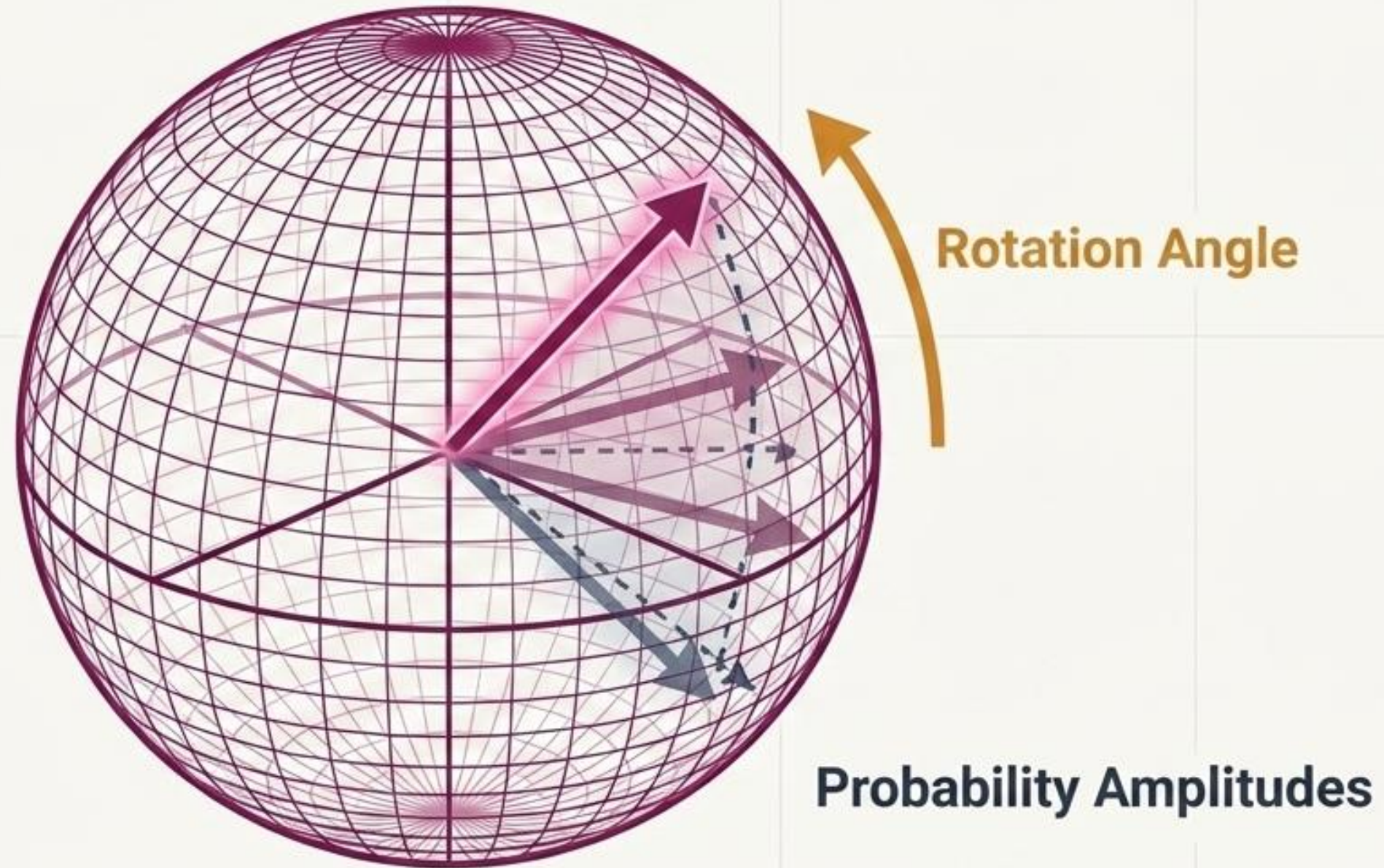
Goal: System-wide efficiency, economic dispatch, and V2G scheduling.

Architecture: Quantum-Inspired Reinforcement Learning (QIRL).



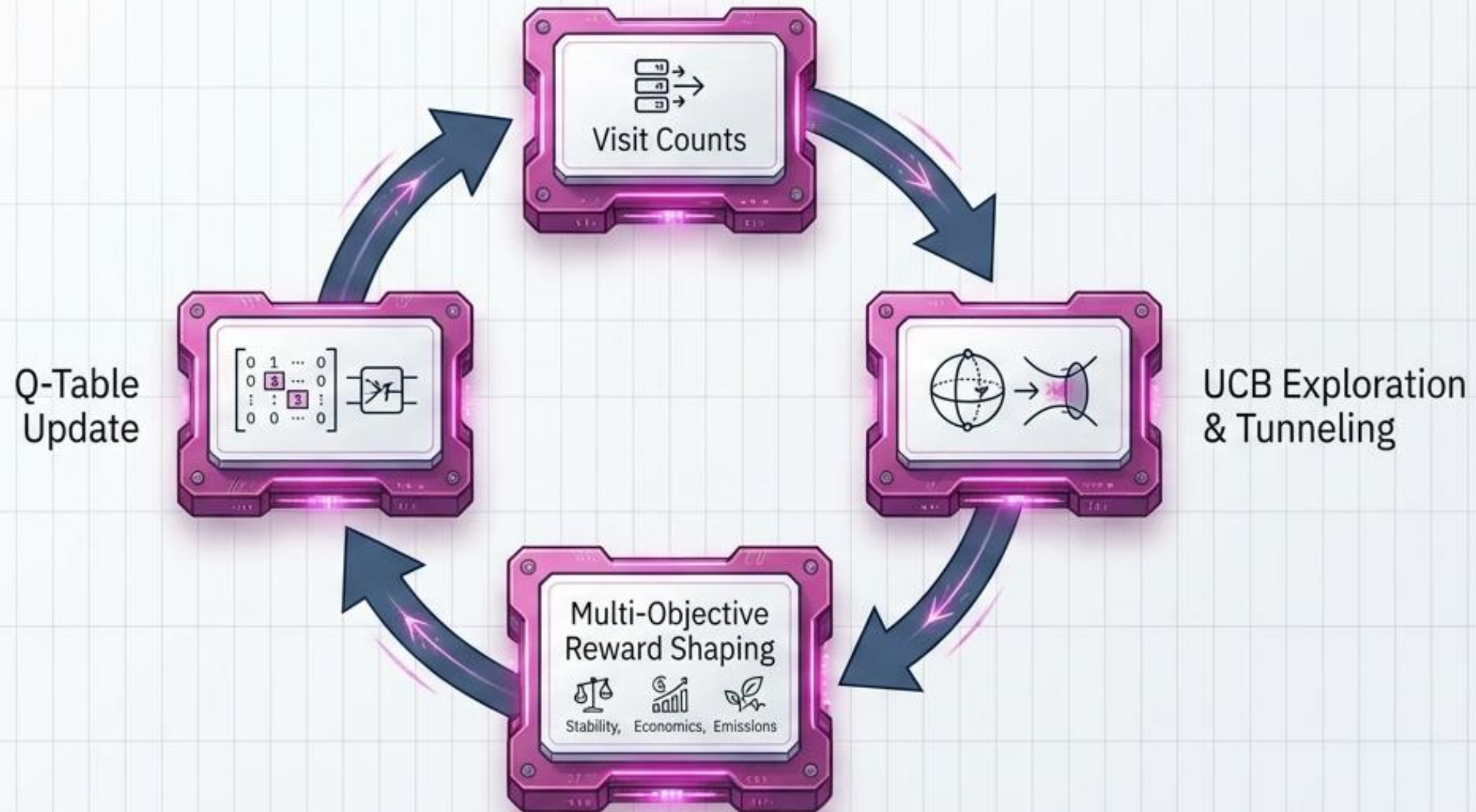
System 2: Quantum-Inspired Representation

- **State Representation:** Uses rotation angles as complex-valued probability amplitudes.
- **Superposition:** Evaluates multiple overlapping control actions simultaneously rather than testing sequentially.



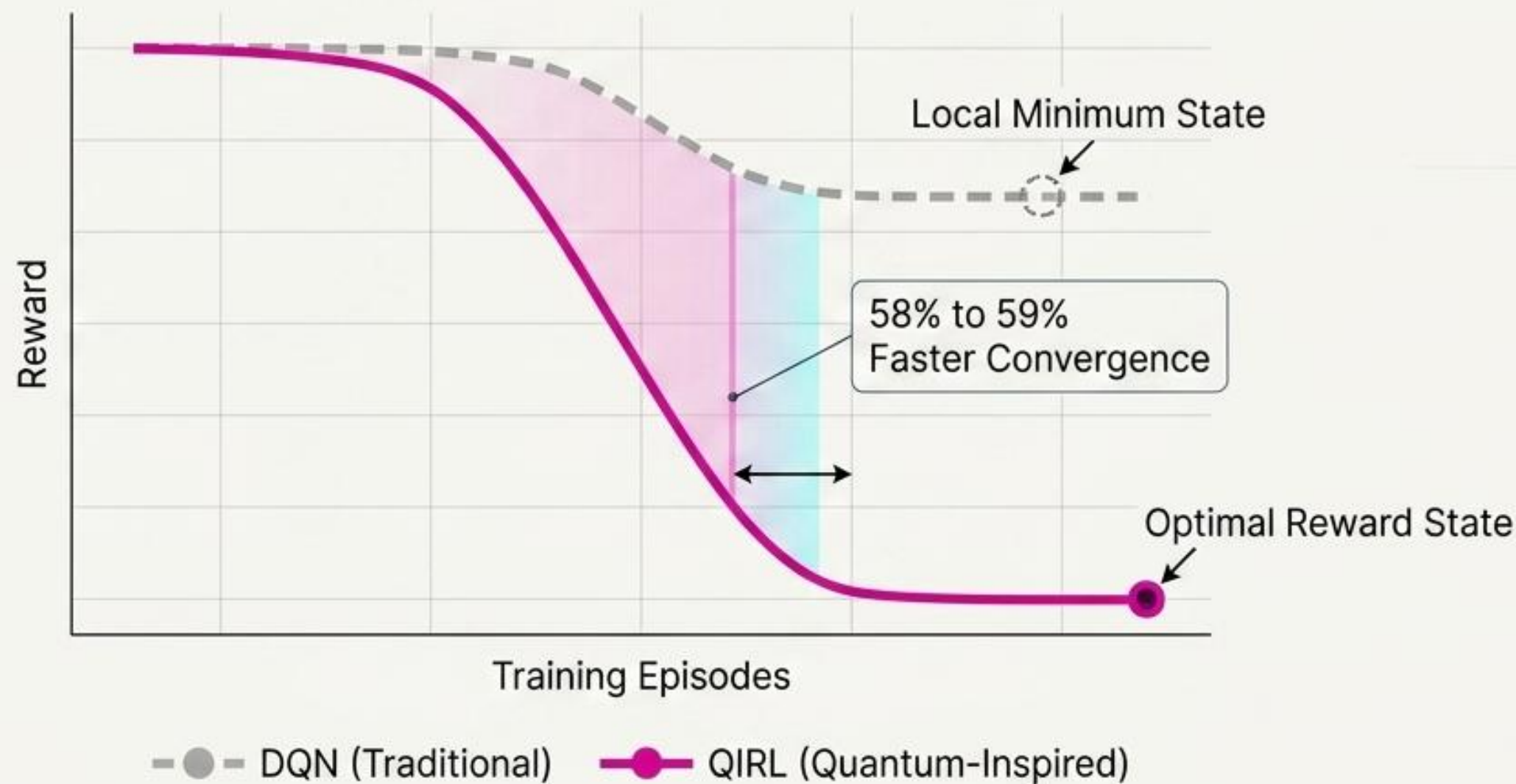
System 2: QIRL Algorithm Details

Exploration uses Upper Confidence Bound (UCB) combined with quantum tunneling to escape local optima



QIRL vs. DQN Convergence Superiority

- **Result:** 58% to 59% faster convergence compared to traditional Deep Q-Networks.
- **Impact:** Crucial for real-time grid adaptation where operating conditions change by the minute.



The Executive Layer: Metacognitive Arbiter Overview

- **Role:** The Prefrontal Cortex of the CAPSM framework.
- **Function:** Decides which system controls the physical grid at any exact microsecond based on internal and external context.



Four Metacognitive Arbitration Modes

SYSTEM_1_REFLEX

Fast response for immediate faults.

SYSTEM_2_PLANNING

Complex, steady-state economic decisions.

SYSTEM_2_OVERRIDE

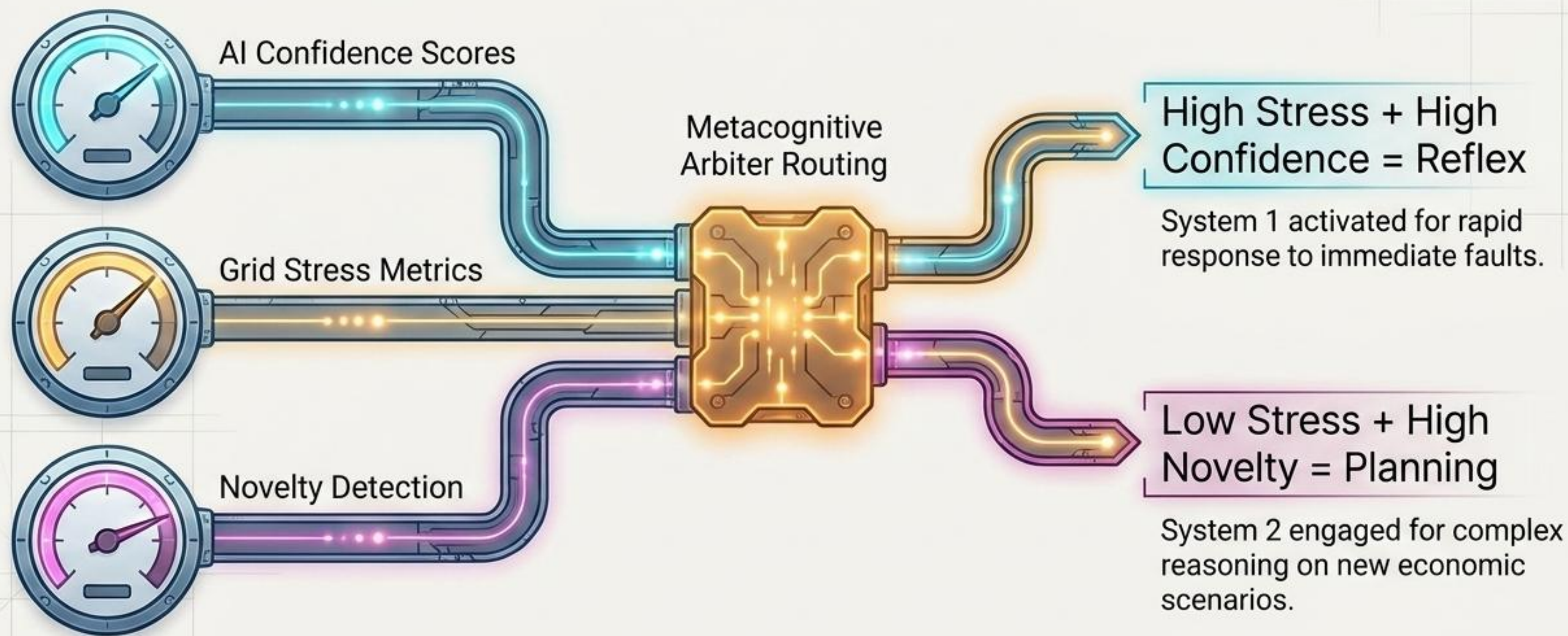
Triggered when System 1 confidence drops on novel data.

SAFE_MODE

Fallback for severe cyber-attacks or critical emergencies.

Dynamic Mode Selection Logic

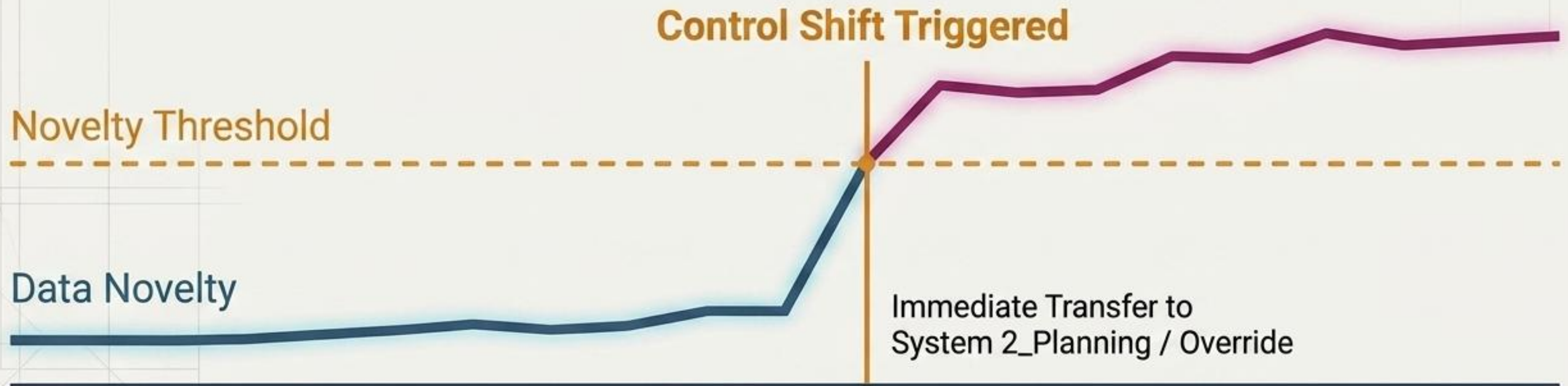
Core Principle: The Arbiter continuously evaluates confidence against urgency.



Override Conditions and Transition Thresholds

Trigger: Mathematical boundaries detect when the fast neural net encounters unmapped data.

Action: Immediate transfer of control from the fast reflex to the deliberative optimizer.



Safe Mode Fallback & Graceful Degradation

Scenario: False Data Injection (FDI) or extreme, unrecoverable uncertainty.

Action: Disables AI optimization to ensure the lights stay on during a cyber-attack.

Protocol: Reverts to purely physics-based, conservative voltage support.

AI Optimization Disabled

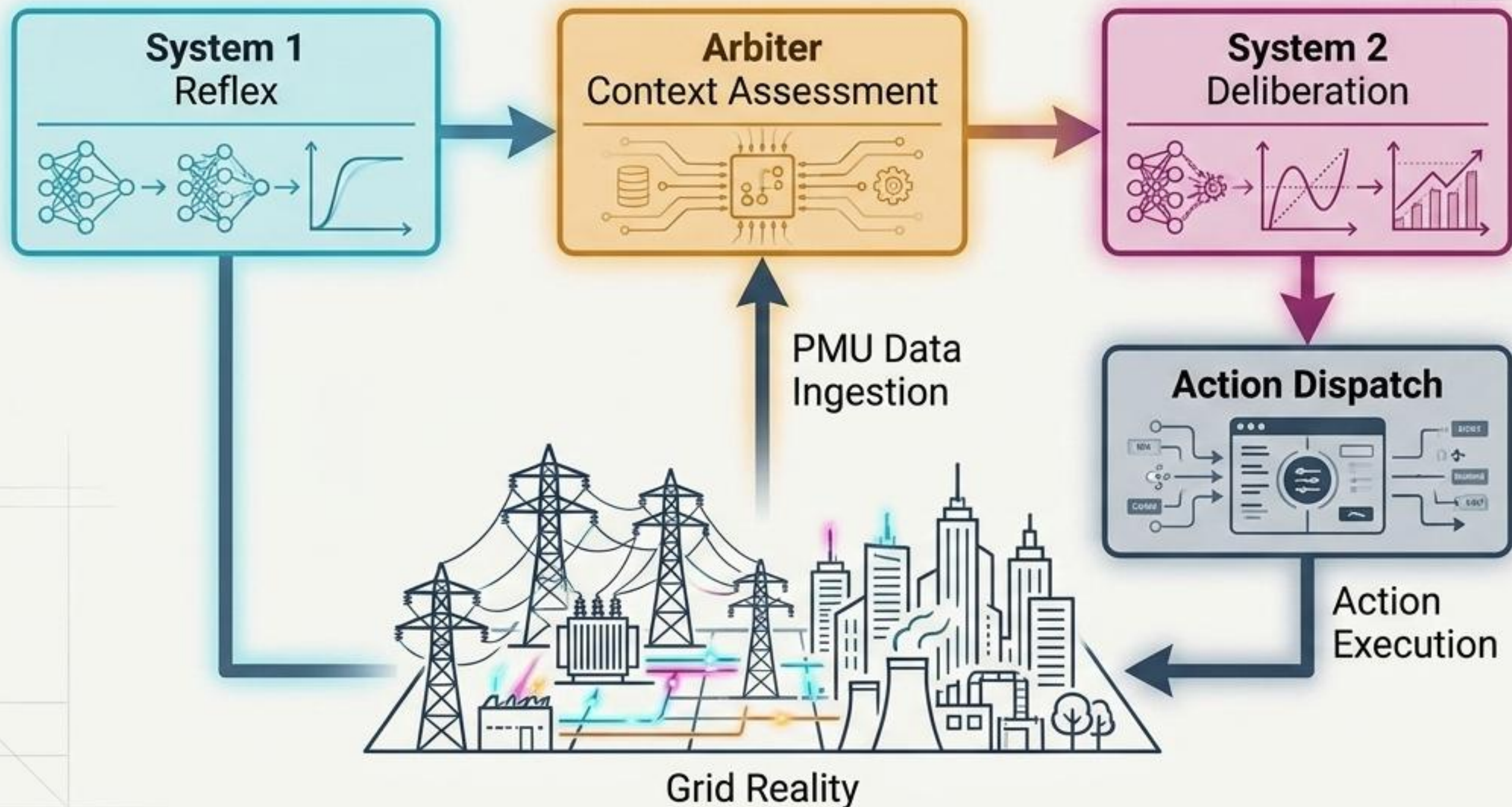


Physics-Based Defense Active



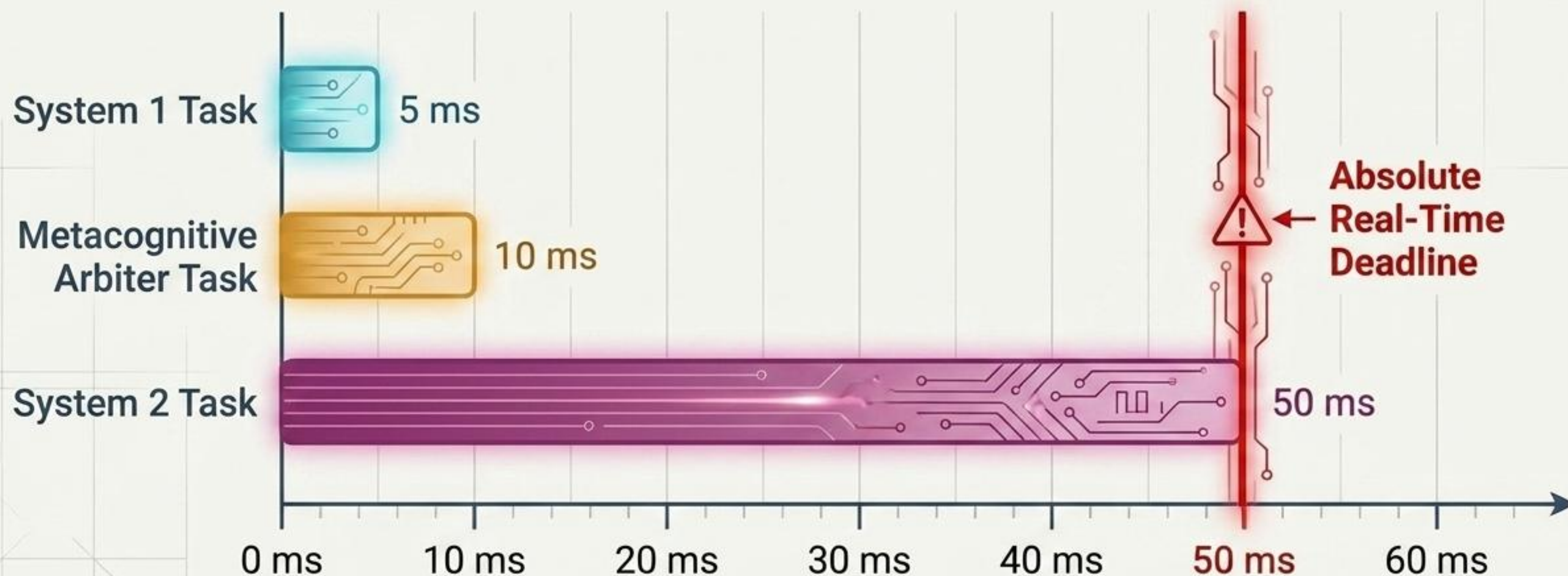
Complete Integration Architecture

All components run in a continuous, multi-rate real-time loop.



Managing Strict Real-Time Constraint Budgets

Requirement: Highly parallelized architecture prevents slow optimization from blocking critical protection responses.



Hardware-in-the-Loop (HIL) OPAL-RT Deployment

Conclusion: CAPSM maps perfectly to real-time silicon for actual utility deployment.

